Done

Building a many-to-many roster

Host: pg.pg4e.com

Port: 5432

Database: pg4e_2165bd5291

User: pg4e_2165bd5291

Password: pg4e_p_4e15809518c8d6 (hide/show copy)

Check Answer

Here is the the **psql** command to access your database:

```
psql -h pg.pg4e.com -p 5432 -U pg4e_2165bd5291 pg4e_2165bd5291
```

Create the following tables.

```
CREATE TABLE student (
    id SERIAL,
    name VARCHAR(128) UNIQUE,
    PRIMARY KEY(id)
);

DROP TABLE course CASCADE;
CREATE TABLE course (
    id SERIAL,
    title VARCHAR(128) UNIQUE,
    PRIMARY KEY(id)
);

DROP TABLE roster CASCADE;
CREATE TABLE roster (
    id SERIAL,
    student_id INTEGER REFERENCES student(id) ON DELETE CASCADE,
    course_id INTEGER REFERENCES course(id) ON DELETE CASCADE,
    role INTEGER,
    UNIQUE(student_id, course_id),
    PRIMARY KEY (id)
);
```

Course Data

You will normalize the following data (each user gets different data), and insert the following data items into your database, creating and linking all the foreign keys properly. Encode instructor with a role of 1 and a learner with a role of 0.

```
Freia, si106, Instructor
Chaela, si106, Learner
Eubh, si106, Learner
Ilsa, si106, Learner
Mahdiya, si106, Learner
Aysha, si110, Instructor
Cesare, si110, Learner
Mathu, si110, Learner
Olubanke, si110, Learner
Takua, si110, Learner
Zainedin, si206, Instructor
Capree, si206, Learner
Carolyn, si206, Learner
Inka, si206, Learner
Luciana, si206, Learner
```

You can test to see if your data has been entered properly with the following SQL statement.

```
SELECT student.name, course.title, roster.role
    FROM student
    JOIN roster ON student.id = roster.student_id
    JOIN course ON roster.course_id = course.id
    ORDER BY course.title, roster.role DESC, student.name;
```

The order of the data and number of rows that comes back from this query should be the same as above. There should be no missing or extra data in your query.